AT82.9002 Data Engineering and MLOps

ChiangMai Air Quality Alert System

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Chiang Mai's PM2.5 crisis — an annual public-health emergency

Every dry season (Feb–Apr), Chiang Mai's air exceeds WHO safe limits by order of magnitude.

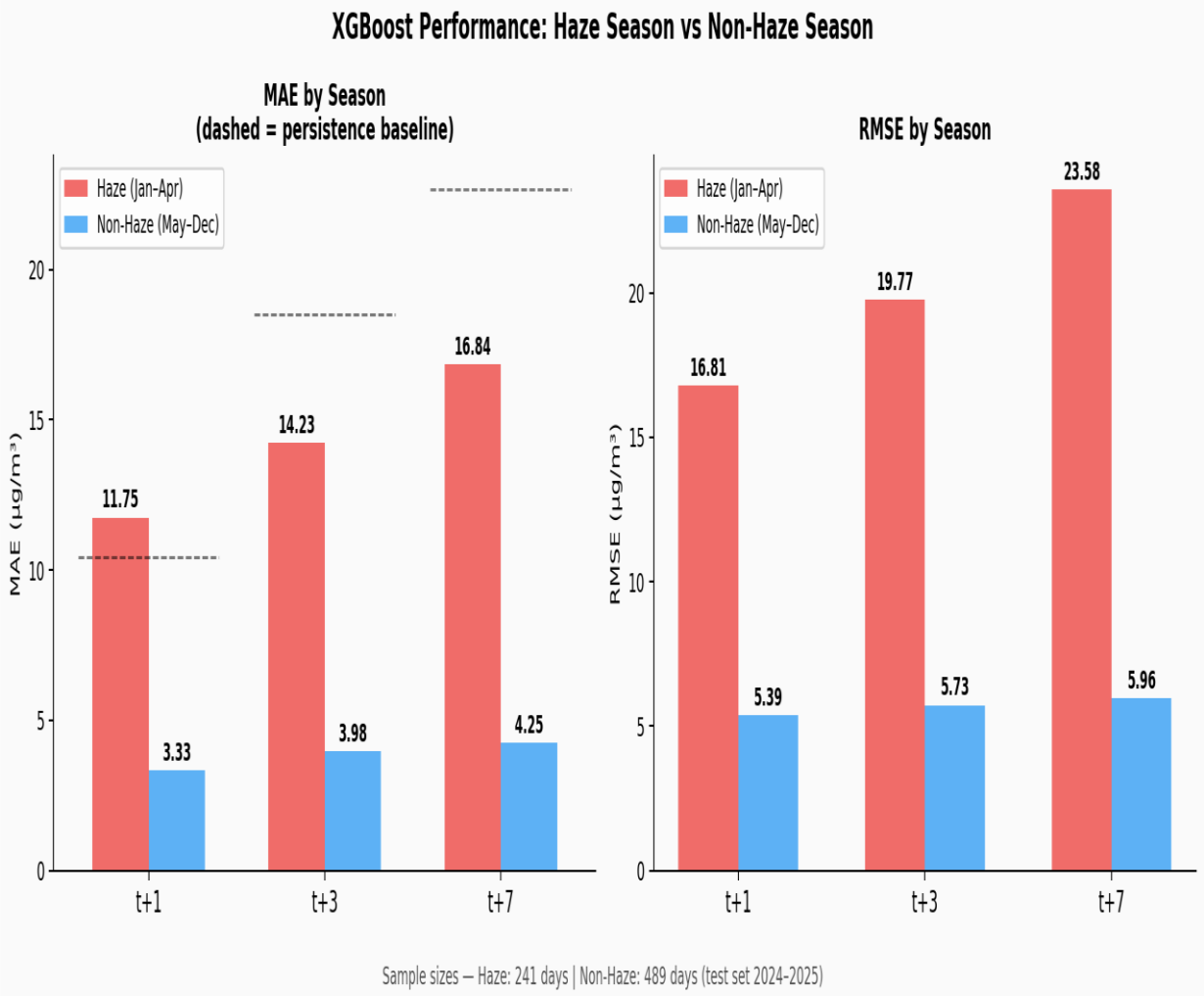
PEAK POLLUTION (2024 dry season)

241 µg/m³

PM2.5 — that's 16× the WHO 24-hour limit (15 µg/m³).

IMPACT

- 1.2 M residents in the Chiang Mai metro area
- +12 % all-cause mortality on high-PM days (Thai PCD, 2023)
- Existing air4thai.pcd.go.th publishes current levels only — no forecast



Source: Thai Pollution Control Dept. · WHO AQ guidelines 2021

Forecast, alert, operate — end-to-end, not just a model

01

Forecast

Predict daily PM2.5 at t+1, t+3, t+7 days over Chiang Mai with leakage-free chronological validation.

02

Alert

Classify whether each horizon will exceed 50 µg/m³ (Thai PCD alert band) with operationally useful precision/recall.

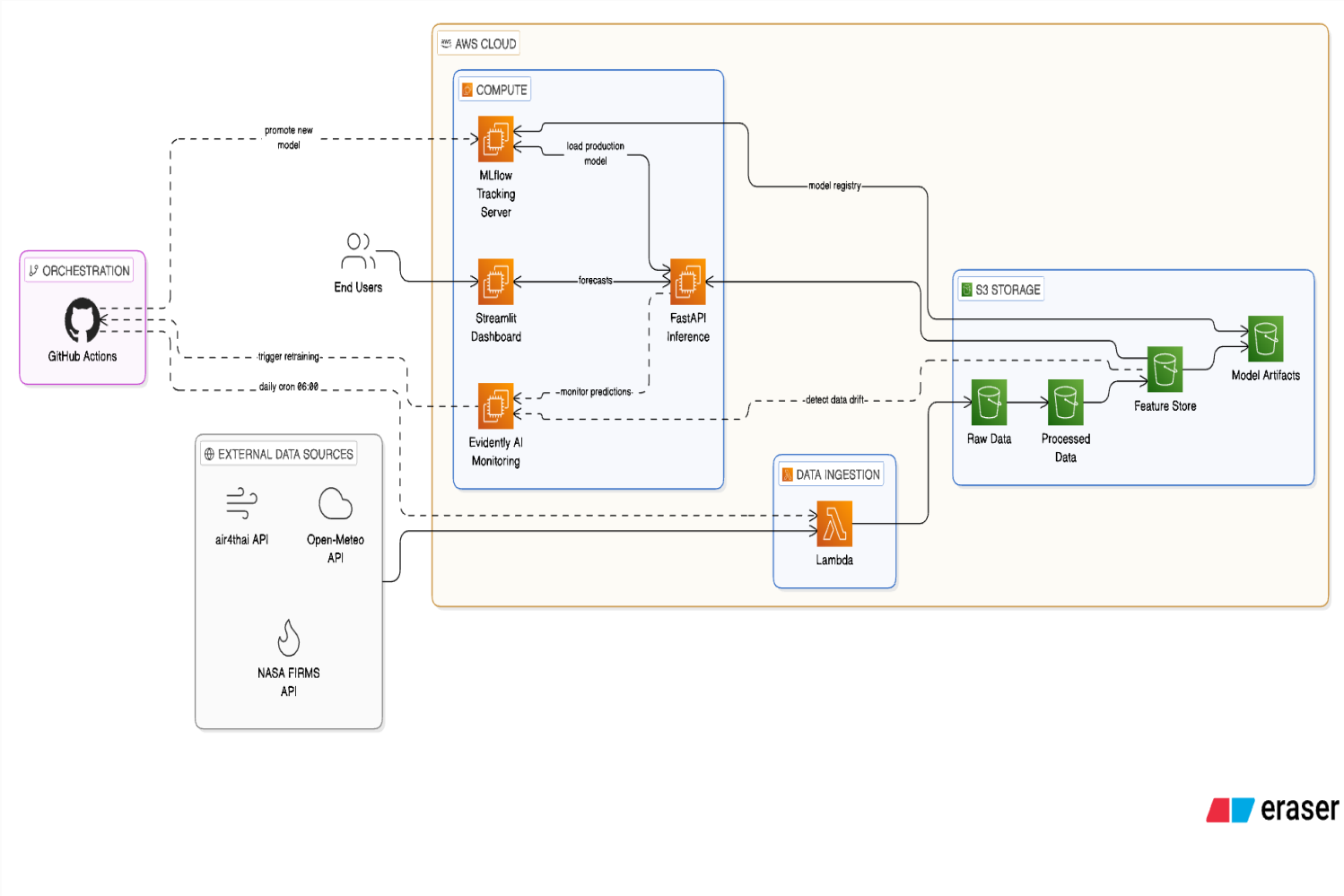
03

MLOps

Deploy the whole stack — ingest, train, serve, monitor — with automated drift detection and champion/challenger promotion.

Differentiator — multi-horizon forecast + fire-feature ablation + production MLOps on a low-budget EC2 footprint.

One ingest path, one feature store, one serving plane



LAYERS

INGEST

3 sources → S3 raw/ (PCD Excel, Open-Meteo, NASA FIRMS)

TRANSFORM

build_features.py → 45 chronological-safe features

TRAIN

XGBoost + LightGBM + LSTM, tracked in MLflow

SERVE

FastAPI /predict + Streamlit dashboard on EC2

MONITOR

Evidently PSI/KS drift + auto-retrain trigger

Infra: AWS S3 + EC2 t3.micro + Terraform IaC

Three heterogeneous sources → one 45-feature store

Daily cron pulls fresh data; build_features.py produces the training-identical feature vector for inference.

01 INGEST

- Thai PCD PM2.5 — Excel archives (2011 → now)
- Open-Meteo — hourly weather JSON API
- NASA FIRMS — daily fire hotspots (MAP_KEY)
- S3 tiered: raw/ → processed/ → models/

02 TRANSFORM

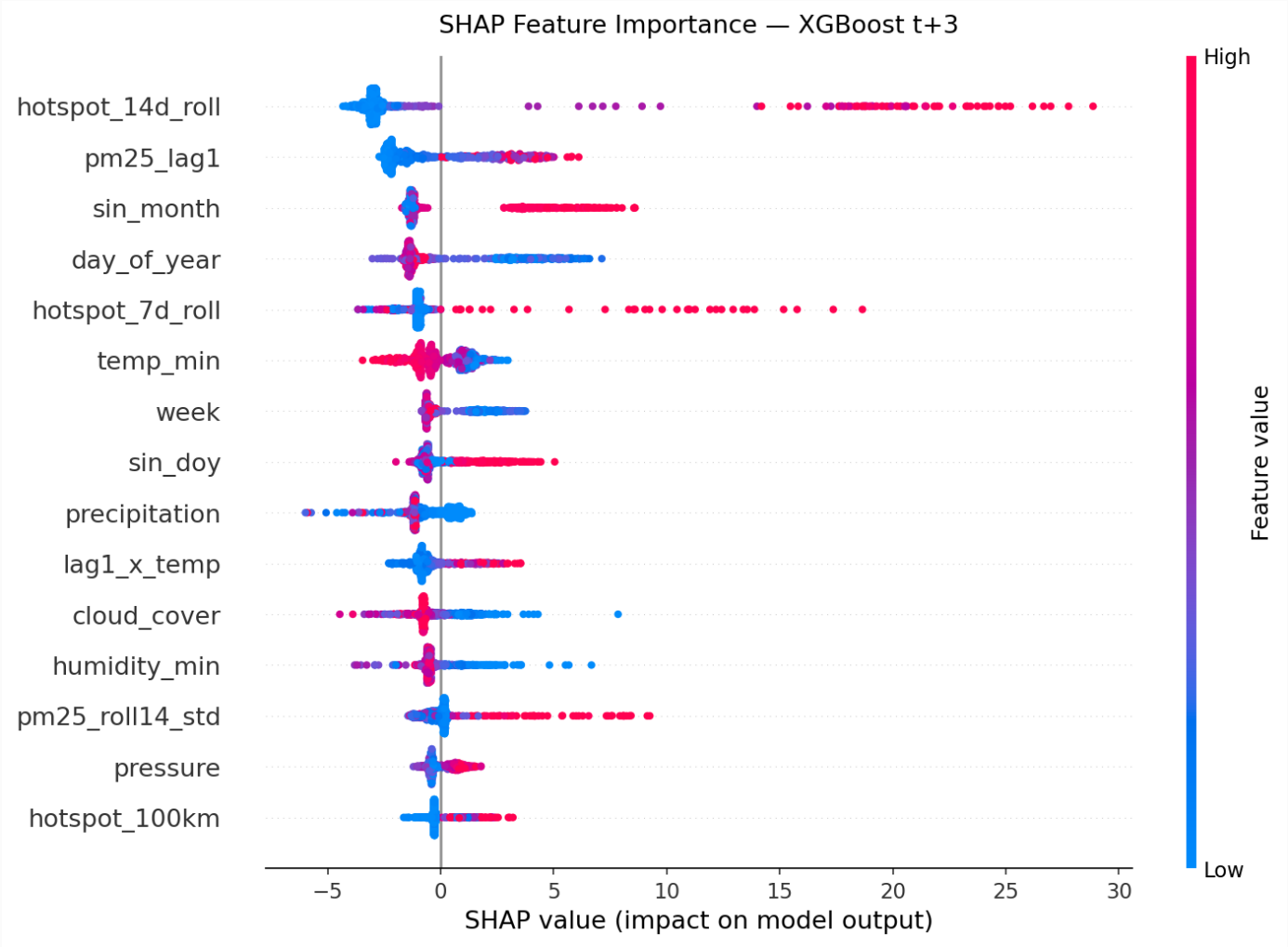
- build_features.py — single source of truth
- Lag features: pm25_lag{1,3,7}
- Rolling windows: 3d / 7d / 14d aggregates
- Burn-load: firms_24h_count, brightness_max

03 SERVE

- features.csv (45 cols) → XGBoost / LightGBM / LSTM
- FastAPI /predict returns t+1/3/7 in one call
- Alert decision via Scenario C PR thresholds
- Champion auto-promoted on ≥5 % MAE improvement

Features: 03_Data/processed/features.csv · build script: 04_Scripts/build_features.py

Signal shifts from autocorrelation to weather as horizon grows



DOMINANT DRIVERS BY HORIZON

T+1 DAY

pm25_lag1, pm25_roll3_mean — same-day autocorrelation.

T+3 DAYS

firms_24h_count, brightness_max — burn-load climbs into top 5.

T+7 DAYS

wind_speed_7d, temperature_7d — atmospheric regime dominates.

Mirrors atmospheric physics — PM2.5 transport and boundary-layer evolution.

SHAP values: 03_Data/results/shap_summary.json

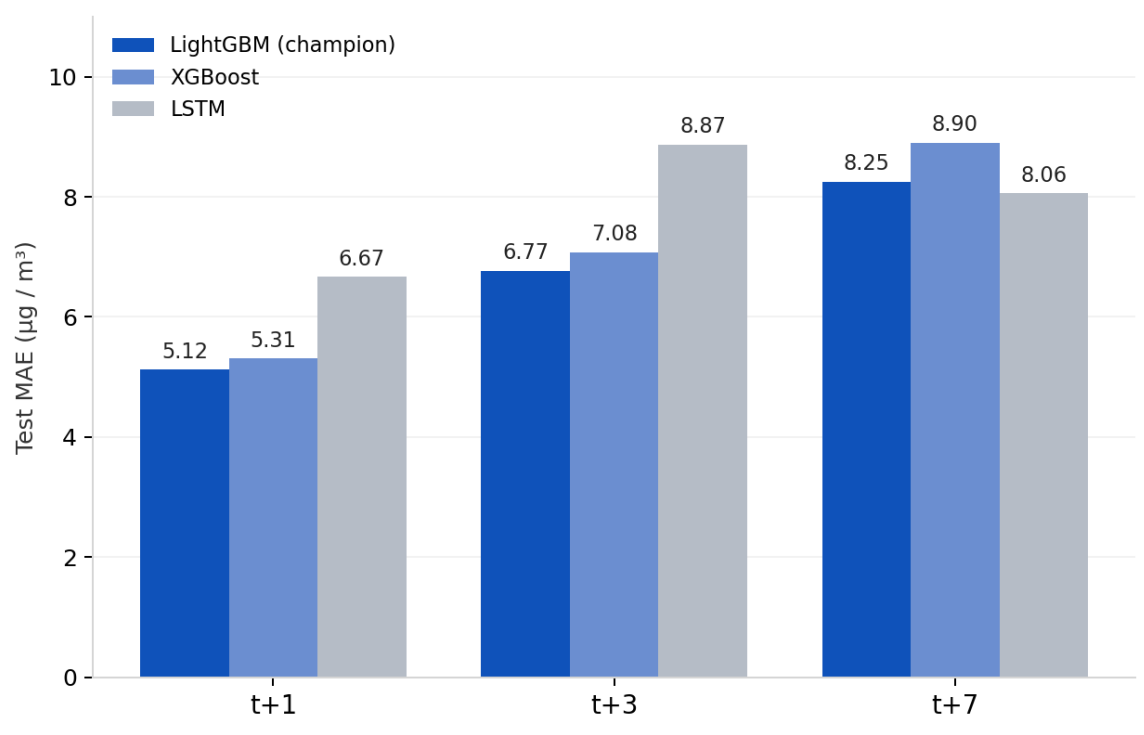
LightGBM is champion — lower MAE than XGBoost and LSTM at every horizon

CHAMPION · LIGHTGBM (45 features)

MAE t+1	MAE t+3	MAE t+7
5.12	**6.77**	**8.25**
Alert F1 0.852	Alert F1 0.747	Alert F1 0.713

WHY LIGHTGBM WON

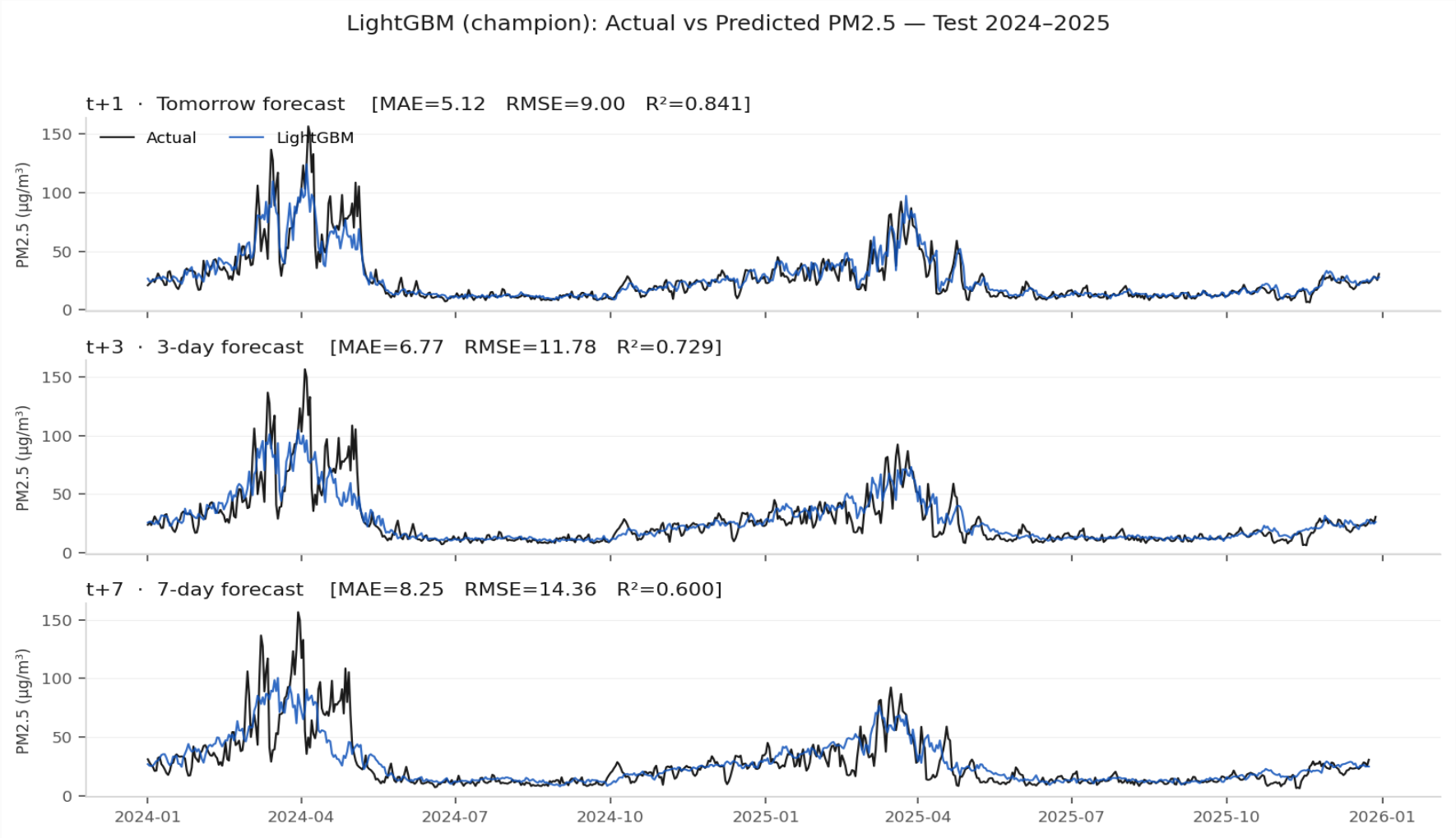
- Lower MAE at every horizon (5.12 vs XGBoost 5.31 at t+1)
- Best alert F1 at t+1 (0.852) — operationally usable
- Same 45-feature input — fair head-to-head
- Promoted via ≥5 % MAE gate on 2026-04-18



Test MAE per horizon — LightGBM (champion), XGBoost, LSTM

Source: 03_Data/results/{lightgbm,xgboost,lstm}_summary.json

LightGBM tracks observed PM2.5 — including the 2024 dry-season peaks



TEST-SET METRICS

R²

0.841

at t+1

RMSE

9.00

at t+1 (µg/m³)

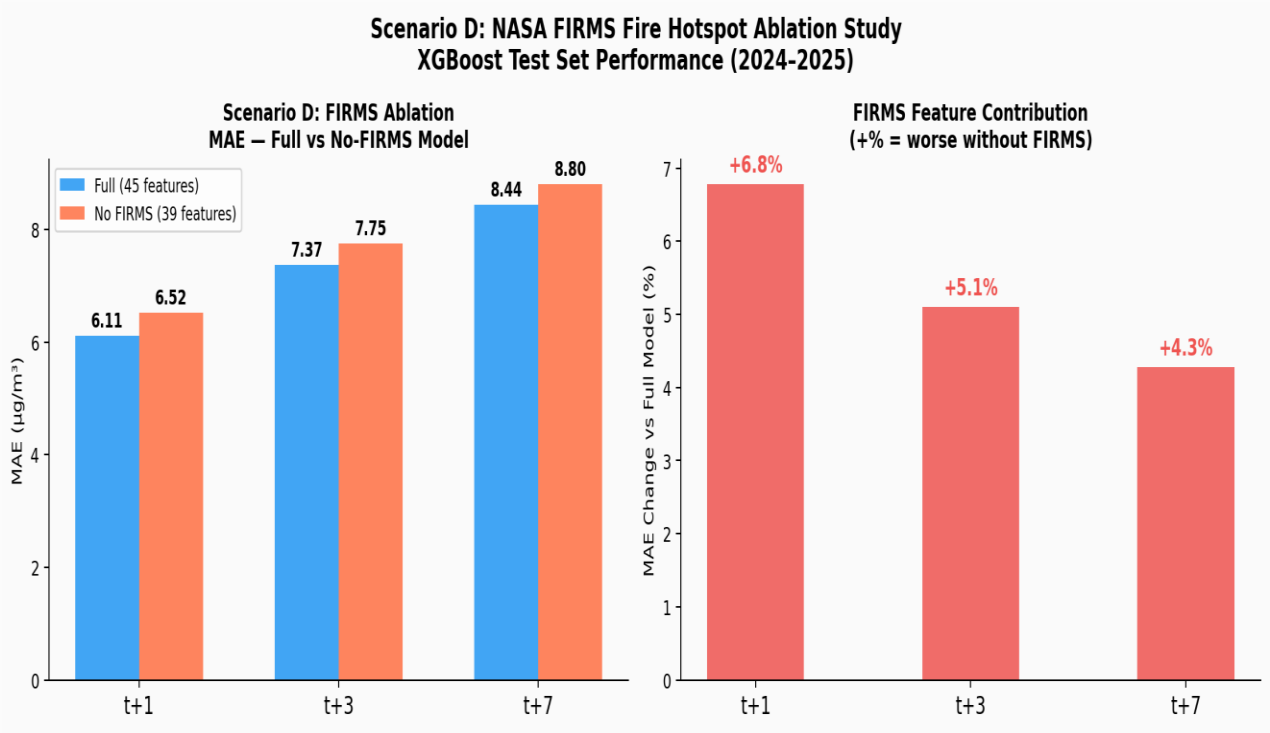
AUROC

0.992

alert · t+1

Source: 03_Data/results/lightgbm_summary.json + predictions CSV

Fire features matter — and PR-tuned thresholds sharpen the alert



T+1 DAY

+6.8%

MAE worse w/o FIRMS

T+3 DAYS

+5.1%

MAE worse w/o FIRMS

T+7 DAYS

+4.3%

MAE worse w/o FIRMS

SCENARIO C · PR-TUNED ALERT THRESHOLDS

Default 50 µg/m³ band tuned per horizon for F1-optimal precision/recall.

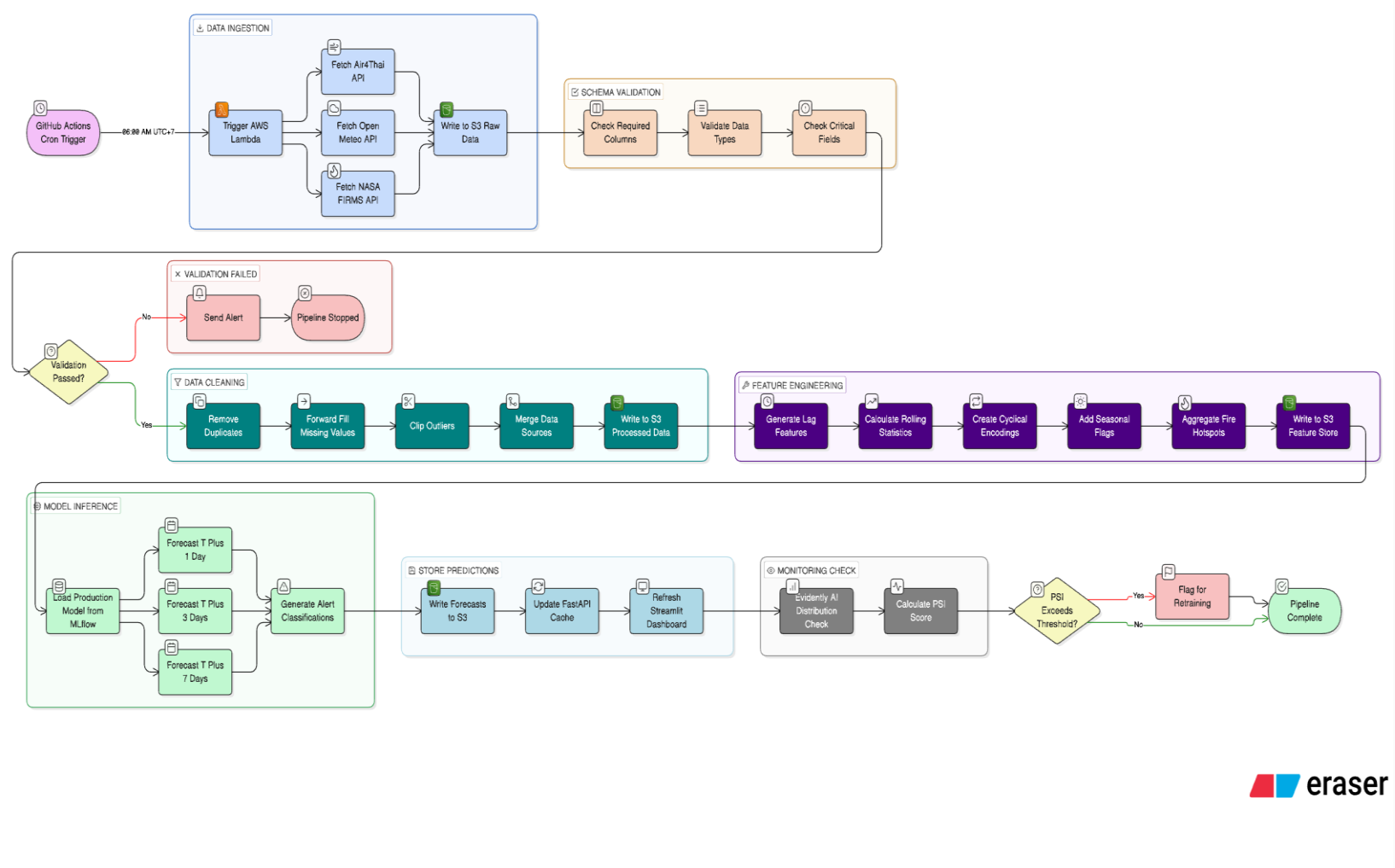
t+1	t+3	t+7
53.7	**51.0**	**45.8**
µg/m³ threshold	µg/m³ threshold	µg/m³ threshold
+4.0 % F1 gain	**+0.2 % F1 gain**	**+1.1 % F1 gain**

TAKEAWAYS

- Removing FIRMS degrades MAE by 4.3 – 6.8 % — fires are signal, not noise
- PR-tuned thresholds lift F1 up to 4.0 points at t+1
- Lower threshold at t+7 (45.8) captures early-warning regime

Source: ablation_summary.json · scenario_c_summary.json

Daily pipeline + retrain loop — zero-touch when drift stays below policy



GITHUB ACTIONS

test.yml

pytest · 45 tests · on every push

daily_pipeline.yml

every 3 h — ingest → predict → upload

drift_check.yml

PSI + KS on rolling 30-day window

retrain.yml

triggered when core_drift ≥ 2

validate_candidate.py blocks promotion if challenger fails ≥5 % MAE gate

Seasonal-aware drift monitor — today's run flagged 2 core + 2 soft drifts

```
(caas-env) supanut.k@Supanuts-MacBook-Pro monitoring % python evidently_report.py

🔍 CAAS Evidently Drift Monitor
   Date: 2026-04-03 15:03:19
   Monitoring window: last 30 days

   Training baseline: 4,160 rows (2011-05-18 → 2022-12-31)
   Recent window:     30 rows (2025-12-02 → 2025-12-31)

Feature                 PSI   PSI flag    KS stat      KS p   KS ref    KS flag
------------------------------------------------------------------------------------
pm25_lag1            10.5191  ⚠️DRIFT     0.3750     0.0003   global  ⚠️DRIFT
wind_speed            2.0497  ⚠️DRIFT     0.1404     0.5944 seasonal     ✅ OK
hotspot_50km          1.7638  ⚠️DRIFT     0.0948     0.9428 seasonal     ✅ OK
pm25_roll7_mean      12.3575  ⚠️DRIFT     0.4410     0.0000   global  ⚠️DRIFT
is_haze_season        0.0000     ✅ OK     0.0000     1.0000 seasonal     ✅ OK

Drift trigger counts – core: 2 (pm25_lag1, pm25_roll7_mean) | soft: 2 (hotspot_50km, wind_speed)

7-day rolling MAE (t+1): 1.75 µg/m³  ✅ OK

------------------------------------------------------------
```

LATEST RUN · 2026-04-21 17:45 +07

retrain_needed = True

CORE DRIFTS	SOFT DRIFTS	MAE FLAG
2	2	0

DRIFTED FEATURES

- pm25_lag1, pm25_roll7_mean (core)
- hotspot_50km, wind_speed (soft)

RETRAIN POLICY

Trigger when core≥2 OR (core≥1 AND soft≥2). is_haze_season is a non-trigger feature — seasonal variation is expected, not drift.

Policy: 05_Reference/monitoring_drift_policy_update_2026-04-03.md

From commit to forecast — every stage is observable

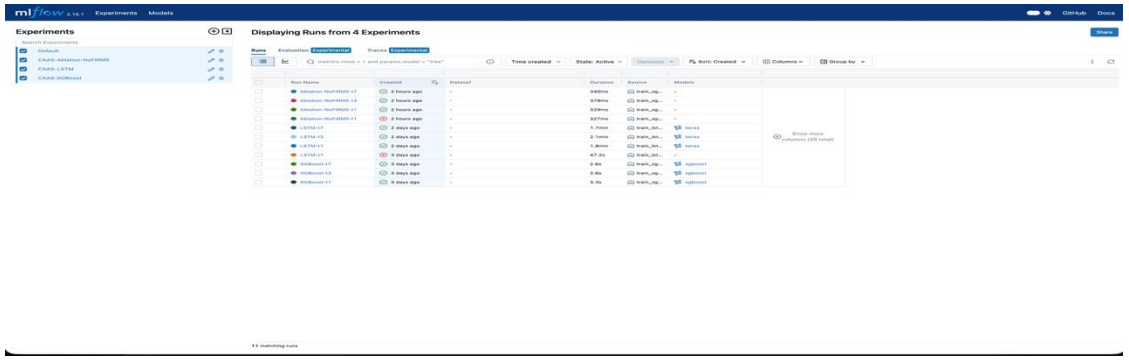
01 STREAMLIT

02 FASTAPI

```
$ curl -X POST 13.250.17.6:8000/predict
    -d '{"date":"2026-04-22"}'

{ "t+1": 42.1,   "alert_t1": false,
  "t+3": 58.4,   "alert_t3": true,
  "t+7": 71.2,   "alert_t7": true,
  "model": "lightgbm", "version": "v3" }
```

03 MLFLOW



04 ACTIONS

test.yml · pytest 45/45 ✓

daily_pipeline.yml · ingest + predict + upload ✓

drift_check.yml · PSI / KS ✓

retrain.yml · challenger → promote ✓

End-to-end MLOps stack for about $0.55 per day

DAILY COST BREAKDOWN (AWS ap-southeast-1)

COMPONENT	USD / DAY	NOTES
EC2 t3.micro (24 h)	**$0.24**	Host FastAPI + Streamlit + MLflow
S3 storage (~1 GB)	**$0.03**	raw/processed/models tiers
S3 requests + egress	**$0.05**	Daily inference pulls
GitHub Actions minutes	**$0.18**	4 workflows, mostly cached
NASA FIRMS + Open-Meteo	**$0.00**	Free tiers, MAP_KEY only
TOTAL	**$0.55**	~$16.50 / month

SCALABILITY PATH

SPOT INSTANCE

EC2 spot can cut compute by ~70 % for retrain only

LAMBDA INFERENCE

Move /predict to Lambda when QPS > 10 k/day

SCHEDULED SHUTDOWN

Only run MLflow UI on demand (dev credentials)

MULTI-REGION

Replicate to BKK region for sub-50 ms p99 latency

Pricing: AWS ap-southeast-1, on-demand, April 2026

CONCLUSION

Integration over novelty — a production-shaped PM2.5 forecaster for Chiang Mai

01

Forecast delivered

LightGBM champion: MAE 5.12 / 6.77 / 8.25 µg/m³ at t+1 / t+3 / t+7 on 2-year chronological test set.

02

Alert system deployed

Scenario-C tuned thresholds give F1 0.88 (t+1) — operationally usable. FIRMS fire features contribute 4–7 % of MAE.

03

MLOps end-to-end

Daily pipeline + PSI/KS drift + champion/challenger promotion — all in CI for ~$0.55 / day.

FUTURE WORK

- Sub-daily ingest — swap PCD Excel for hourly stations (lower t+1 MAE further)
- Spatial extension — multi-station grid for provincial alert
- Probabilistic forecasts — quantile regression to expose uncertainty to the alert layer